Quiz — 1.4.2 Data Structures

Name: ____ Date: _ ***Mark:*** ___ / 30

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Section A — Multiple choice (8 marks)

1. Which data structure operates on a **Last In, First Out** basis? A. Queue B. Stack C. Linked list D. Binary search tree Your answer: ☐

2. Which statement best describes a **record**? A. An ordered collection of items of the same type accessed by index B. A collection of named fields that may hold different data types C. An immutable ordered sequence D. A set of nodes each pointing to the next Your answer: ☐

3. Which is the key property of a **static** data structure? A. It grows and shrinks at run time B. Its size is fixed when it is declared C. It is always stored on the heap D. It cannot be accessed by index Your answer: ☐

4. In a circular queue of size 8, the rear pointer is updated using which expression? A. `rear = rear + 8` B. `rear = (rear + 1) MOD 8` C. `rear = rear − 1` D. `rear = (rear − 1) MOD 8` Your answer: ☐

5. Which traversal of a binary search tree visits the nodes in **ascending order**? A. Pre-order B. Post-order C. In-order D. Level-order Your answer: ☐

6. What is a **collision** in a hash table? A. When the table becomes full B. When two different keys hash to the same index C. When a key cannot be hashed at all D. When the load factor reaches zero Your answer: ☐

7. A major disadvantage of a **linked list** compared with an array is that it: A. Cannot grow at run time B. Must store items of different types C. Has no random access — it must be traversed from the head D. Uses no extra memory Your answer: ☐

8. Which describes the correct ends used by a **queue**? A. Enqueue at the front, dequeue at the rear B. Enqueue and dequeue both at the top C. Enqueue at the rear, dequeue at the front D. Enqueue at the rear, dequeue at the rear Your answer: ☐

Section B — Calculations & short answer (17 marks)

9. Match each description to the correct data structure (Array, Tuple, Linked list, Hash table, Stack). **[5]**

#	Description	Structure
a	Ordered, **immutable** sequence such as `(x, y)`	_____
b	LIFO structure with push/pop at the top	_____
c	Fixed-size, same-type collection accessed by a numeric index	_____
d	Nodes of data + a pointer to the next node	_____
e	Maps a key to an index using a hash function for ~O(1) lookup	_____
		(Linked list)

10. A **stack** is initially empty. The following operations are carried out in order. Trace the stack and give the values popped and the final contents. **[4]**

`push(5)`, `push(9)`, `push(2)`, `pop()`, `push(7)`, `pop()`

Working space (show the stack after each operation):

Values popped (in order): _ ***Final stack (top → bottom):*** _

11. A **queue** uses `enqueue` (rear) and `dequeue` (front). Starting empty, perform: **[3]**

`enqueue(A)`, `enqueue(B)`, `dequeue()`, `enqueue(C)`, `enqueue(D)`, `dequeue()`

Working space:

Values dequeued (in order): _ ***Queue contents (front → rear):*** _

12. The following numbers are inserted, in this order, into an empty **binary search tree**: **50, 30, 70, 20, 40, 60**. Draw the resulting tree. **[3]**

Working space:

13. State **two** differences between a **static** and a **dynamic** data structure. **[2]**

Working space:

Section C — Exam-style question (5 marks)

14. A web browser stores the list of pages the user has visited so the **Back** button can return to the previous page. Items are also held in a buffer that streams video, releasing frames in the order they arrived.

(a) Name the most appropriate data structure for the **Back button** history and justify your choice. **[2]**

(b) Name the most appropriate data structure for the **video buffer** and justify your choice. **[2]**

(c) State one reason a programmer might choose a **dynamic** implementation for either structure. **[1]**

Working space: